

OpenRAM Overview

> Relevant source files

OpenRAM is an open-source static random access memory (SRAM) compiler framework written in Python. It generates SRAM layouts, netlists, timing and power models, placement and routing models, and other views necessary for ASIC design integration. This document provides a high-level overview of OpenRAM's purpose, architecture, and capabilities. For detailed instructions on installation and usage, see [Installation and Setup](#) and [Usage and Basic Workflow](#).

Core Capabilities

OpenRAM provides a complete framework for generating custom SRAM memories with the following capabilities:

- Generation of physical layouts (GDSII format)
- Creation of circuit netlists (SPICE format)
- Timing and power characterization
- Liberty file (.lib) generation for STA tools
- Verilog behavioral models
- LEF abstracts for place & route
- HTML datasheets with specifications
- Design rule checking (DRC) and layout vs. schematic (LVS) verification

Sources: [README.md](#) 16-20 [setup.py](#) 60-96

Supported Technologies

OpenRAM works with multiple technologies through technology-specific configuration files:

Technology	Description	Location
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FreePDK45	Open-source 45nm process	technology/freepdk45
SCMOS  Fast 	Scalable CMOS	technology/scn4m_subm 

Technology	Description	Location
Sky130	SkyWater 130nm open PDK	technology/sky130

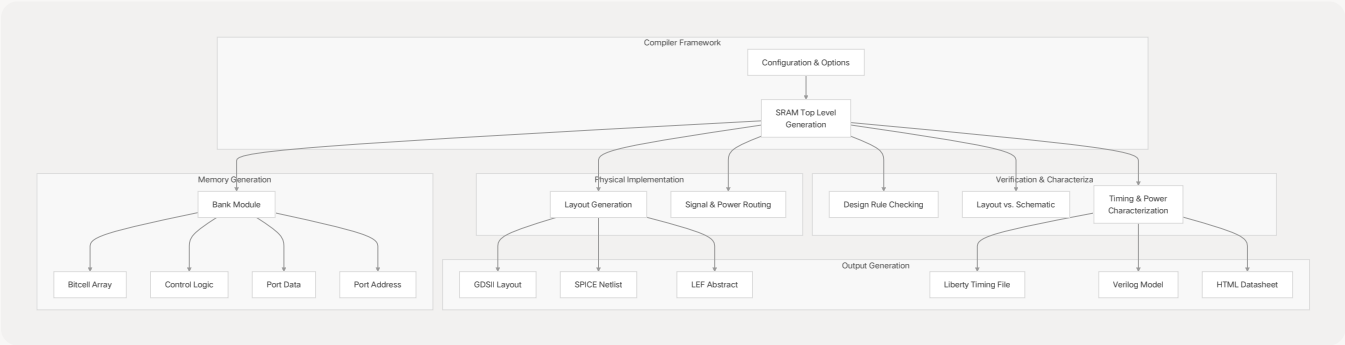
Users can add support for new technologies by creating appropriate technology files following the porting guidelines.

Sources: [README.md](#) [104-106](#) [CONTRIBUTING.md](#) [22-26](#)

System Architecture

OpenRAM implements a modular, hierarchical design that separates different aspects of memory generation.

High-Level Architecture

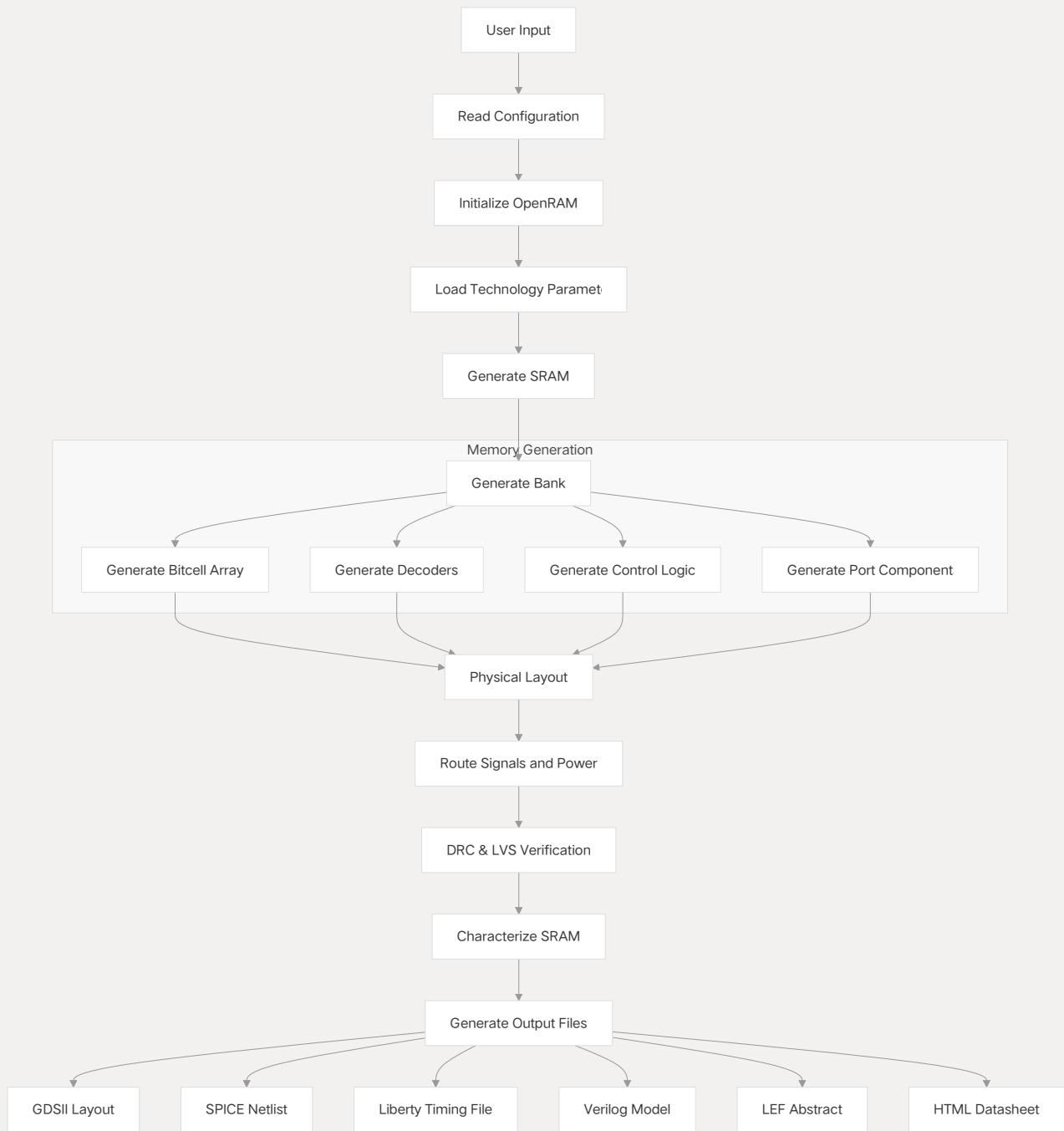


Sources: [__init__.py](#) [44-50](#) [CONTRIBUTING.md](#) [10-29](#)

Compilation Flow

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Sources: `__init__.py` 44-50 `compiler/verify/run_script.py` 11-92

Key Components

OpenRAM is organized around several key Python modules that implement different aspects of

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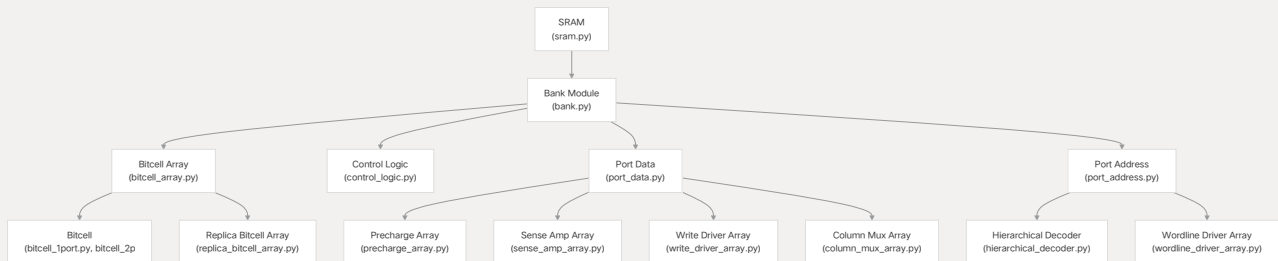
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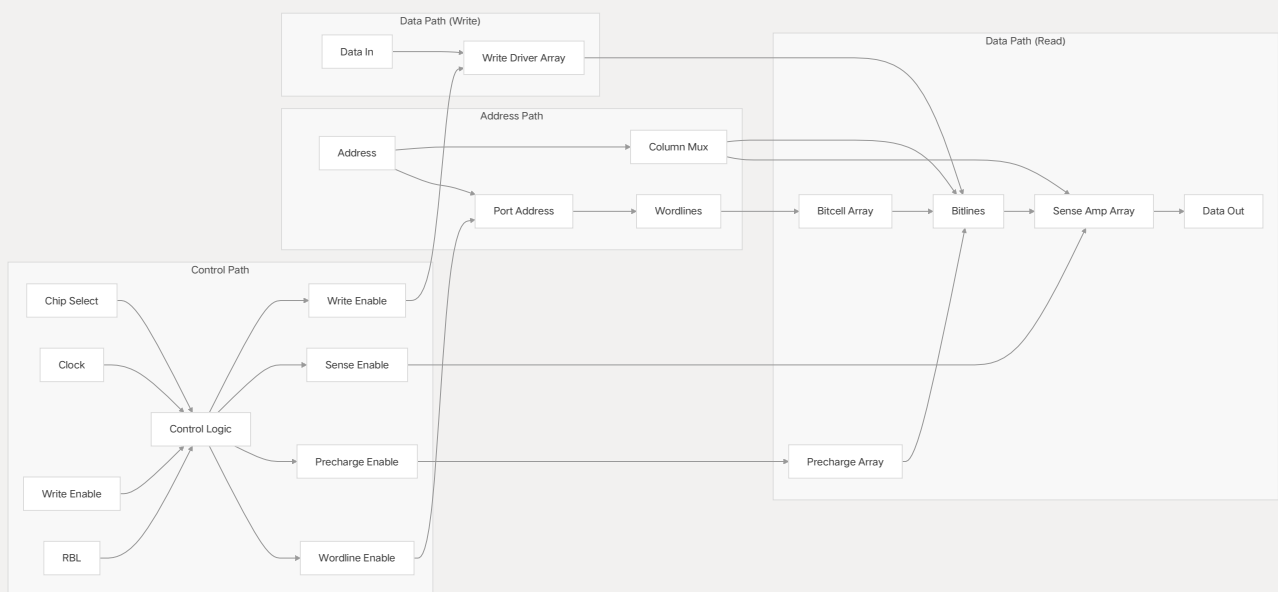
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Sources: `compiler/modules/__init__.py` 6-87

Data and Control Flow



Sources: `compiler/modules/__init__.py` 6-87

Physical Design System

OpenRAM includes a sophisticated physical design system that handles layout generation and routing:

1. **Hierarchical Layout System:** Generates layout for all components in a bottom-up approach, starting with primitive cells and building up to the full SRAM.
2. **Router:** Handles connections between components with several specialized sub-routers:
 - Graph Router: Routes signals using maze routing algorithms

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- Supply Router: Routes power and ground networks



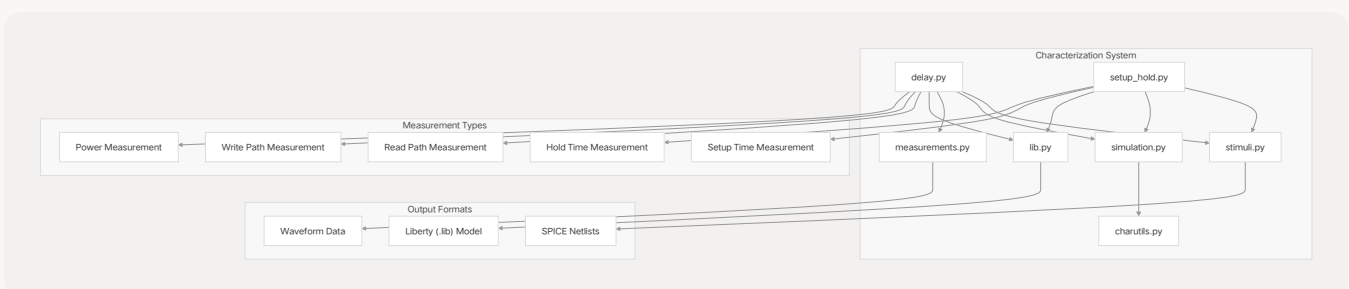
3. **Verification:** Interfaces with external tools for Design Rule Checking (DRC) and Layout vs. Schematic (LVS) verification:

- Magic: Open-source layout tool for DRC/LVS
- Calibre: Commercial verification tool
- Netgen: Open-source LVS tool

Sources: `CONTRIBUTING.md` 10-29 `compiler/verify/run_script.py` 11-92

Characterization System

The characterization system generates timing and power models for the SRAM:



The characterization system measures:

- Read and write delays
- Setup and hold times
- Power consumption (active, standby, leakage)
- Minimum operating voltage

Results are compiled into a Liberty (.lib) timing file for integration with synthesis and place & route tools.

Sources: `compiler/verify/run_script.py` 11-92

ROM Support

Besides SRAM, OpenRAM also supports Read-Only Memory (ROM) generation. The ROM configuration works similarly to SRAM but with predefined data:

- Users can specify ROM data in binary or hex format

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- Supports various word sizes and organizations
- Generates similar output formats to SRAM (GDSII, SPICE, Liberty, etc.)

For example, a 1KB ROM can be configured with:

```
word_size = 1
rom_data = "rom_configs/example_1kbyte.bin"
data_type = "bin"
```

Sources: [compiler/rom_config.py](#) 16-146 [macros/rom_configs/sky130_rom_1kbyte.py](#) 1-20

Version and Development

OpenRAM is under active development. The current version is documented in the VERSION file:

1.2.48

Development follows open-source contribution guidelines, with code available on GitHub. Users can contribute by:

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